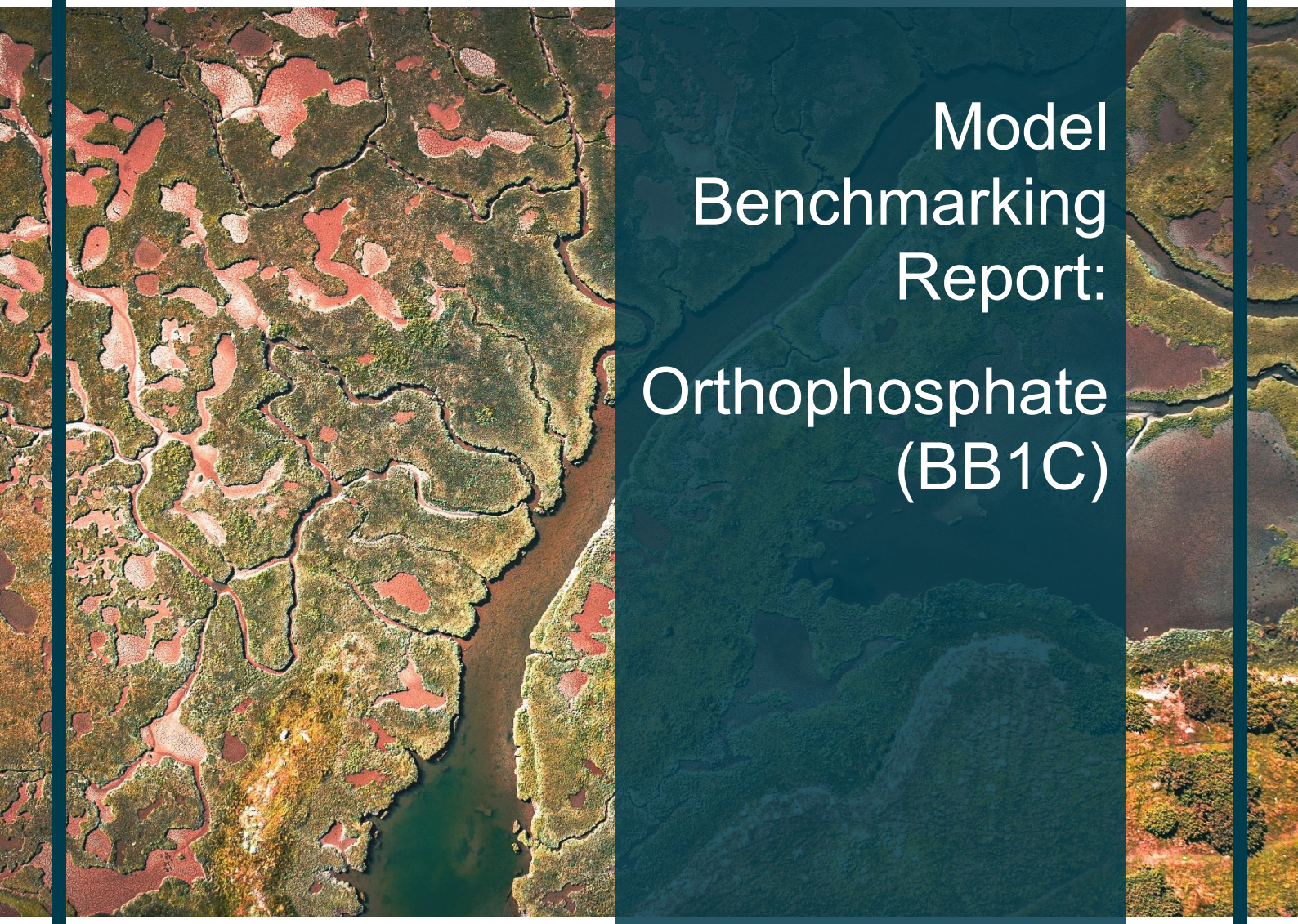
River Deep Mountain AI



Model Benchmarking Report: Orthophosphate (BB1C)

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January 2026

Model Benchmarking Report - Orthophosphate (BB1C)

FINAL REPORT

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Glossary

AI	Artificial Intelligence
BOD	Biochemical Oxygen Demand
CL	Confidence Limit
EA	Environment Agency
INCA-P	INtegrated CAtchment model of phosphorus
LightGBM	Light Gradient Boosting Machine
ML	Machine Learning
NRMSE	Normalised Root Mean Squared Error
PP	Particulate Phosphorus
R^2	Coefficient of Determination
RDMAI	River Deep Mountain AI
RMSE	Root Mean Squared Error
RRMSE	Relative Root Mean Square Error
SAGIS	Source Apportionment Geographical Information Systems
SHAP	Shapley Additive exPlanations
SRP	Soluble Reactive Phosphorus
SS	Suspended Sediments
TDP	Total Dissolved Phosphorus
TP	Total Phosphorus
UKWIR	United Kingdom Water Industry Research
WFD	Water Framework Directive

Summary

Model Outputs	SAGIS	INCA-P	RDMAI Model
Model Approach	SAGIS (Source Apportionment Geographical Information Systems) provides predictions of river water quality based on pollution source databases. The river quality model SIMCAT is used to predict annual flow and water quality statistics as annual means and percentiles in river catchments by including estimates of river flow and chemical input distributions to river reaches for a set of 38 pollutants. The SIMCAT predictions are derived on a stochastic basis using a combined distributions Monte-Carlo approach. The statistical results can be compared to water quality standards and reported as a compliance assessment.	The INtegrated CAtchment model of phosphorus (INCA-P) is part of a suite of models representing different water quality compounds. INCA-P is a process-based, mass-balance model that dynamically simulates phosphorus processes in both the plant-soil system and in-stream components of river catchments. The model also uses spatial information on sub-catchments and land use. INCA-P is a semi-distributed model, i.e. it simplifies the physical catchment into land-use classes within each sub-catchment and simulates the processes within each class separately, instead of accounting for the full spatial distribution of the land features. This reduces the data and computational requirements. The model outputs daily time series and summary statistics of flow and water quality for the simulated sites.	The RDMAI Open OrthoP is based on an Artificial Intelligence (AI)/Machine Learning (ML) framework using historical observations from the Environment Agency (EA) water quality archive. The model is built using LightGBM (Light Gradient Boosting Machine), a high-performance algorithm based on gradient boosting decision trees, well-suited to capture non-linear relationships in data. The model uses SHAP (Shapley Additive exPlanations) to explain the model predictions by quantifying the contribution of each input feature to the model results. SHAP is a key tool to ensure transparency and interpretability of ML models, as part of Responsible AI requirements.

Model Outputs	SAGIS	INCA-P	RDMAI Model
Applications	The models' primary use is assessing the impact of effluent permit changes/applications on water quality. Other uses include source apportionment for water company planning, providing information on background water quality, reasons for failure of water quality standards for each river catchment, and available headroom to inform investment plans.	The model is used to investigate day-to-day variations in flow and water quality following changes in the input conditions (e.g. temperature and precipitation) as well as changes in land use. Applications include management of water pollution, evaluation of the effectiveness of different land management and pollution control strategies and studying the potential impacts of climate change and land-use changes on water quality.	The model helps understand the relationships between the water quality determinands in the EA database and the orthophosphate concentration in rivers, providing orthophosphate concentration estimates, including for sites where orthophosphate is not currently monitored. This provides additional information to aid and supplement water quality monitoring and assessment.
Geographic Constraints	SAGIS is built around UK-specific datasets and therefore is suitable for studies in England, Wales and Scotland.	INCA-P was developed for the UK and Europe, but has been applied to studies in Asia, Africa and the Americas, and therefore is it not limited to a geographical location. However, it requires that the catchments are suitable for subdivision into semi-distributed reaches with homogeneous land use, soil, slope and hydrology.	The RDMAI Open OrthoP was developed and validated using sampling point data measured in EA's operational catchments in England, therefore currently more suited to these geographical areas.
Temporal Resolution	SAGIS provides a long-term annual average concentration for 38 pollutants including phosphate. Results include source	INCA-P is designed to provide daily variations in flow and water quality at selected points along the river. Other outputs include profiles of flow or water quality at selected times, cumulative frequency distributions of flow and water quality, tables	The granularity of the output depends on the frequency of measurements of the determinands used as model inputs. This means that if the input determinands are available at a daily rate, the estimates of orthophosphate will also be provided at a

Model Outputs	SAGIS	INCA-P	RDMAI Model
	apportionment, confidence limits around the mean and percentiles.	of model performance statistics, and daily and annual pollution loads for land uses and processes.	daily rate, even if the orthophosphate measurements are less frequent or not present for a particular site.
Data Inputs Required	The source apportionment data and model structures are already set up for the user. There are instructions on editing the model structure and inputs as required. This involves setting up summary statistics (e.g. mean phosphate for wastewater treatment works) or seasonal distributions (e.g. monthly diffuse phosphate loads from agriculture). The data updates are limited to periodic updates coordinated around the 5-year water company investment cycle and the EA requirements to use SAGIS results for the 6-year River Basin Planning and reporting cycle. The user cannot update all the load estimates as these require inputs	Input requirements include time series data of precipitation, air temperature, and hydrologically effective rainfall. Land use, soil properties and topography information are required for subdividing the catchment into sub-catchments and associated river reaches. Spatial data (e.g. reach length and slope) are required to simulate the movement of phosphorus within the river. INCA-P specifically also needs data on the inputs of both diffuse and point-source phosphorus. Input data may be from a mixture of open and licenced sources depending on the user's requirements. Pre-processing of the inputs is required before using it in the model.	The model was developed using 45 determinands available in the EA water quality archive in timeseries form for multiple sampling points. However, the model was developed in such way that it is still possible to estimate the orthophosphate using fewer determinands. Determinands from the EA water quality archive or from other (e.g. private) data collection schemes could be used to run the model. Other input requirements include sample point location (including coordinates), date of the sample, purpose of the sample and where the sample was compliant. The data will need to be pre-processed before using as model inputs.

Model Outputs	SAGIS	INCA-P	RDMAI Model
	from other water quality models and processing other datasets.		
Model Performance	The UKWIR SAGIS model estimates generally showed a positive trend relationship with mean EA measures of orthophosphate P, but overall overestimated measured concentrations. For the points considered, the UKWIR SAGIS model obtained a relative root mean square error (RRMSE) of 512% and an R^2 of 0.05. This benchmarking did not attempt to correlate the UKWIR SAGIS model performance with contributing catchment characteristics given the large number of points involved and the need to delineate the upstream contributing catchment to each point. Given the infrequency of EA measures, spikes in orthophosphate are likely missed. Therefore, the higher estimates by the UKWIR SAGIS model are not necessarily incorrect.	Literature indicates that INCA-P tends to provide reasonable, useful results. For the catchment/monitoring point used for benchmarking, the model performed generally well for the different parameters simulated in both calibration and validation stages. Simulating daily total dissolved phosphorus (TDP) and particulate phosphorus (PP) dynamics is challenging in catchments where diffuse phosphorus sources dominate. In common with the majority of daily models, the model therefore struggles to achieve positive Nash-Sutcliffe Efficiency values, which would indicate a good model performance. However, the model dynamics are plausible, with concentrations of the right magnitude during baseflow, increasing by the right order around rainfall events. While an analysis directly on the soluble reactive phosphorus (SRP), or orthophosphate, is not provided in this study, the results of the TDP are an indicative of the model performance in terms of orthophosphate. TDP was available at a higher frequency than SRP for this study and is highly correlated to SRP in the Tarland.	Validation of the model was carried out using selected sampling points (from the EA water quality archive) previously unseen during model development. The EA data used in the validation were measured from June 2024 to July 2025. The model performed better at predicting orthophosphate concentrations under normal conditions, but its predictive accuracy declined at sites with extreme high or low orthophosphate readings. These results suggest that the model's assumptions and training data do not adequately capture the full range of variability present in outlier scenarios. The validation report recommended that the model should be applied with caution to sites known to be experience unusually high or low orthophosphate P concentrations. The model was also validated using EA sonde water quality measurements, compared to citizen science data collected at the same locations. The model struggled at locations where the measured orthophosphate concentrations were high. Benchmarking the model performance against that of an UWKIR SAGIS dataset required a comparison to EA monitoring data on an annual mean basis. With this limitation in mind, the averaged RDMAI Open

Model Outputs	SAGIS	INCA-P	RDMAI Model
			OrthoP model outputs for points assessed were closer to the mean of EA monitoring data than the annual mean produced by the UKWIR SAGIS dataset, with an RRMSE of 80%. However, the RDMAI Open OrthoP model seemed to be relatively insensitive to increases in EA observed orthophosphate P up until a concentration of around 0.1 mg/l, after which the RDMAI estimates increased, reflected by a relatively lower R^2 of −0.22. This indicates that the RDMAI Open OrthoP model may be more applicable to sites of a Moderate or Poor status according to the UTAG standards for orthophosphate P. Benchmarking the model performance against the INCA-P dataset was constrained by the limited number of determinands available and by geographical differences. This highlighted some limitations of the RDMAI Open OrthoP model due to the differences in characteristics of the data used to train the model compared to the INCA-P data. A comparison between the model simulations at the single site, Coull, showed that both models struggled to represent observed orthophosphate-P, particularly at peak concentrations. However, the RDMAI Open OrthoP model tended to over-estimate orthophosphate-P, with a NRMSE of 31.4% compared to that of INCA-P of 14.8%.

Model Outputs	SAGIS	INCA-P	RDMAI Model
Cost & Time	The SIMCAT software and specific catchment model files set up by the EA are free to use. These files would include the source apportionment inputs set up by the EA to represent diffuse and point loads. To run SAGIS, users must purchase the latest UKWIR SAGIS report and hold a license from WHS for LowFlows Enterprise for the specific geographical region of interest. The SAGIS report and tool are available to purchase from UKWIR for a base price, which has been listed at approximately £600 + VAT, plus additional fees for data licensing. The SAGIS platform can only be used by experienced water quality modellers and requires access to ArcGIS Pro and purchase of the SAGIS tools. ArcGIS Pro licence is a named licence, which costs about £4,200/yr. Time is needed to derive water quality baselines for the	The software may be free for academic users, but a paid-for licence is needed for commercial users and consultants. Prices are provided upon enquiry to Water Resources Associates. Some of the data (e.g. climate data) may need to be purchased for commercial use in some countries. The cost will depend on the required datasets for a particular application. Using INCA-P requires intermediate to advanced modelling skills and knowledge of GIS core functions to set it up, including catchment/sub-catchment delineation, land use mapping and attribution of soil properties.	The RDMAI Open OrthoP is open source and free to use. The model was developed using freely available data from the EA water quality archive. Data costs are possible if the model is run using alternative sources of input water quality data, i.e., non- EA water quality data – costs will depend on the data provider. Considerable time (days) is required to process the input data to create additional features and format the data as required for the model. An experienced Python programmer is required to understand and perform the steps for input data preparation and to run the model.

Model Outputs	SAGIS	INCA-P	RDMAI Model
	purpose of using SAGIS. A past WRc project allocated around 4 days for using SAGIS to derive water quality baselines. Using an existing SAGIS dataset, such as the UKWIR SAGIS dataset used in this benchmarking, requires a simple download. Extracting the required data from a pre-existing SAGIS dataset takes around 0.5 days.		
Strengths	• Provides predictions of long-term water quality for a wide range of pollutants across all Water Framework Directive (WFD) river reaches for the UK, that can be easily compared to statistical water quality standards. • Includes estimates of source apportionment for a comprehensive range of sectors, to allow management decisions to be made at catchment and river-basin scale.	• It is process based and provides relatively detailed results, but simpler and with less data requirements than fully distributed models. • Proven track record, having been used for studies worldwide to simulate, e.g. land-use changes and climate change impacts, including future scenarios. • User-friendly Windows-based interface providing easy visualisation of parameters and transparent structure. • Model parameter auto-calibration and uncertainty analysis tools are available.	• The model can estimate orthophosphate concentrations including at sampling points where orthophosphate is currently unmonitored. • Although the model was trained using 45 water quality determinands, it can also achieve relatively good performance with fewer determinands. This was demonstrated in one of the tests carried out during the model development, where the model was applied to data consisting of 11 determinands measured in a single monitoring point to evaluate its performance (*refer to model report*). • The model combines machine learning methods with established domain expertise to make sure

Model Outputs	SAGIS	INCA-P	RDMAI Model
	• Continues to be used routinely for quality discharge permit applications to protect rivers from pressures across sectors. • Alternative Access: Some outputs of the SAGIS-SIMCAT models are made available to water companies and the Environment Agency via the CaBA (Catchment Based Approach) Data Package. • Custodian: The Environment Agency acts as the custodian of the English and Welsh SAGIS models, and after purchasing the report and holding the required LowFlows license, UKWIR provides contact details to download the executables.		that the predictions are physically plausible and to provide actionable insights. • The model incorporates the latest EA water quality data. Due to its open source status, the model can be re-trained using updated EA water quality data as soon as those data are available. • The use of SHAP provides transparency and interpretability, which are critical for regulatory compliance. • The model could be used to 'fill in' missing orthophosphate-P in the EA's water quality database. • The model has the potential to estimate orthophosphate-P from high-frequency Section 82 sensor data. • Wastewater modelling often requires defining of upstream river reach flow and water quality boundary conditions. The RDMAI Open Flow and Ortho-Phosphate models can assist in defining these boundary conditions.
Limitations	• The diffuse load inputs cannot easily be updated by users.	• Model can be difficult to calibrate due to the large number of parameters. While	• The model predicts orthophosphate values (soluble reactive) and not total phosphorus, which

Model Outputs	SAGIS	INCA-P	RDMAI Model
	Relies on periodic updates coordinated across UKWIR, EA and other regulators (NRW/SEPA). This time-lag is a weakness and highlights that SAGIS results are appropriate (as designed) for estimates of long-term water quality at the timescale of the 5-year water company planning cycle and the 6-year river basin planning cycle. • High skill requirements to set up and use the SAGIS system. • The UKWIR SAGIS dataset is updated periodically, resulting in results not reflecting the latest available data.	parameter auto-calibration tools are available, calibration likely remains time consuming, requiring advanced modelling skills and expert judgement, e.g. to set appropriate parameter ranges. • As the model is semi-distributed (i.e. it has a simplified representation of sub-catchments), it provides limited details in the outputs, including spatial and process details. • Results are provided as daily time series, limiting the representation of certain outputs, e.g. it does not provide sub-daily high flow spikes.	includes other forms of phosphorus. This distinction is important when interpreting the model results. • Data for training the model were selected to contain only samples with a maximum orthophosphate concentration of 0.5mg/L (value identified by domain experts to exclude outliers). This means that the model predictions are constrained to this range and should not be used to infer orthophosphate concentrations under conditions that are likely to produce higher orthophosphate levels. • Although the model can perform well with fewer determinands than what it was trained for, all the missing determinands need to be assigned a value of zero, which can affect the model performance. • The model was trained on data from EA's operational catchments, which are restricted to England. Even though the model can be run for other catchments, location information will be restricted, and the model will not recognise catchment characteristics that are different from those included in the training data.

Model Outputs	SAGIS	INCA-P	RDMAI Model
			• The model is trained on EA monitored orthophosphate-P. EA spot samples are of low frequency, typically 1-2 samples per month at a site. Therefore, samples likely miss spikes in orthophosphate related to pollution or weather events. However, this data limitation cannot be avoided as the EA water quality database represents the best available dataset of river water quality for England.

1. Overview of the RDMAI Open OrthoP Model

1.1 Background and aim

Orthophosphate-P concentration in a river is an indicator of eutrophication; therefore, an important water quality variable for regulators. While there have been advances in *in-situ* sensor technologies in recent years allowing high-frequency water quality monitoring, orthophosphate-P remains difficult and costly to measure. Consequently, measures of orthophosphate-P are sparse across space and time.

The aim of the RDMAI Open Orthophosphate (OrthoP) model was to leverage artificial intelligence (AI) and machine learning (ML) to find relationships between orthophosphate-P and other water quality parameters in the Environment Agency's (EA) water quality monitoring database (WIMS), thereby allowing orthophosphate-P to be predicted by the values of other water quality parameters in a water sample. Apart from 'filling in' missing orthophosphate-P in the EA's water quality database, the model has the potential to estimate orthophosphate-P from high-frequency Section 82 sensor data.

1.2 Methodology

The model was trained on 24 years of water quality data for river water in the EA's water quality database. This dataset comprises approximately 70 million observations of around 2,300 water quality variables. Identifying the most important water quality variables for establishing relationships with orthophosphate-P involved advanced data science methods (such as agglomerative analysis with Morgan Fingerprints for molecular similarity and K-means++ model for clustering), as well as domain expert input. Within model development, five different ML models were assessed, including statistical, deep learning, tree-based algorithms and gradient boosting models. The final model chosen was a gradient boosting model that consistently gave low residual errors, namely the Light Gradient Boosting Machine (LightGBM) model.

1.3 Model validation

Model validation against unseen EA water quality data (June 2024 to July 2025) obtained a Normalised Root Mean Square Error (NRMSE) of 10.15%. The validation results showed in general good agreement between model simulations and EA data, with some slight under-estimation at higher concentrations. Using high-frequency EA sonde data as model input and validation against citizen science measures of orthophosphate-P indicated that the RDMAI Open OrthoP model struggles to represent high observations of orthophosphate-P.

1.4 Model limitations

Identified limitations of the RDMAI Open OrthoP model include the use of a 0.5 mg/L threshold for removing orthophosphate-P outliers in the EA water quality monitoring database. This threshold is being revisited to assess whether a higher threshold can improve model performance. In addition, the low frequency of the EA monitored data must be recognised as a

limitation. Such low frequency data is likely to miss spikes in orthophosphate-P associated with ephemeral pollution events. However, this limitation is unavoidable as the EA water quality database remains the best available water quality dataset for England with which to train a model.

More information on the model can be found on GitHub.

2. Benchmarking against UKWIR SAGIS

2.1 Overview of SAGIS

SAGIS (Source Apportionment Geographical Information Systems) provides predictions of river water quality based on pollution source databases. The river quality model SIMCAT is used to predict annual flow and water quality statistics as annual means and percentiles in river catchments by including estimates of river flow and chemical input distributions to river reaches for a set of 38 pollutants. Simple river chemistry processes of temperature-dependent decay (applicable to phosphate) and interactions between Biochemical Oxygen Demand (BOD), Ammonia and Dissolved Oxygen are included. Results are presented as long-term predictions that can be interpreted as annual mean river flow and quality, with percentiles and monthly mean values.

The GIS element includes the spatial structure of WFD river network, water bodies and input locations. The chemical source databases hold georeferenced loading data that are assigned to point (wastewater, industry) and diffuse sources (air deposition, agriculture, urban runoff, highways, mining etc.) and allocated as point or diffuse inputs for the rivers within a catchment area polygon. The loading contribution from the sectors and water quality predictions are reported along each river reach, catchment and at river basin district level.

SAGIS is widely used by the UK environmental regulators as a catchment planning tool for source apportionment to assess the relative inputs to rivers from wastewater, agriculture, industry and other sectors. This informs, at a river-catchment scale, the reasons for failure of water quality standards and what actions might need to be taken to improve waterbody health, and helps to inform compliance with the Water Framework Directive (WFD).

The SIMCAT model was designed to be used to set permit levels for wastewater discharges, with the statistical predictions set up to be directly compared to the water quality standards. Confidence intervals based on the number of measurements in the input data are calculated to quantify uncertainty in the predictions. SAGIS is frequently called on to assess the impact of planned changes (developments, wastewater permit changes) on compliance with the environmental quality standards.

As a tool used routinely by regulators, water companies and catchment partnerships, UKWIR coordinates SAGIS development, provides access to it and describes it as:

'SAGIS is, however, most useful conceptualised as a 'system' of interconnected tools and processes made up of mathematical models such as SIMCAT, MPER [Metal permitting] and RQP [single site permitting], the ArcGIS interface, data on discharge and water quality, river network and hydrology data, and external tools that rely on SAGIS input and output data, including calibration tools...' UKWIR, 2025.

The UKWIR website states: ”The system we use to ensure that we always provide the latest version of the SAGIS-SIMCAT model is that the Environment Agency acts as the custodian of all the English and Welsh SAGIS models and databases.” This includes carrying out updates to the model structures to update locations and maintain information about flow and quality inputs. The latest set of updates used data up to 2020 with the tools published by UKWIR[1].

The water quality software tool SIMCAT is freely available under an open-source licence on GitHub, where documentation includes the user manual[2]. The GIS tool (SAGIS ArcPro system) and documentation is available to purchase from UKWIR.

Limitations of SAGIS include data and the outputs. The databases require national-level updates and as they are related to water company investment cycles, the data may not be representative of recent conditions. The load input databases are at a variety of scales, from river reach to catchment area scale. The outputs from SIMCAT are long-term annual statistics that may not predict short-term wet weather pollution.

The SAGIS outputs chosen for the benchmarking were the Source apportionment of nutrient contributions to rivers in England and Wales modelled with SAGIS (accessed November 2025, when the last update was April 2025). The dataset was produced using SAGIS as a UK Water Industry Research Programme (UKWIR) project with support from the Environment Agency and SEPA using data from 2001 to 2012, depending on the data type.

This dataset provides estimates of loads and in-stream concentrations of nitrates and phosphates for England and Wales at a WFD waterbody catchment scale. **Since the RDMAI Open OrthoP Model simulates orthophosphate-P, only the orthophosphate-P data from this SAGIS dataset are of relevance for the benchmarking.** Pollutant inputs from various sectors are represented in this dataset. These include:

- Agriculture: crop and livestock farming. Agricultural phosphates were modelled with PSYCHIC based on 2012 census data
- Highways: Modelled according to the Highways Agency Water Risk Assessment Tool (HAWRAT) based on average annual rainfall, highway network maps, and mean event concentrations reported by the Highways Agency.
- Urban runoff: Calculated according to rainfall intensity, CORINE land use data and mean event concentrations reported in the literature.

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[1] 21/WW/02/14, SAGIS Model Update Tool to Support Modelling for PR24; 21/WW/02/13, UKWIR’s Source Apportionment GIS Model (SAGIS): Version 4.0.

[2] Tony Warn, The SIMCAT Manual, 2018 simcat/docs/181118_SIMCAT_Manual.docx at main · DEFRA/simcat · GitHub

- Wastewater treatment works (WwTWs): calculated using the EA water quality samples database (WIMS) and flow data for WwTWs reported to the EA by water companies (MCERTS - both 2010-2012 inclusive).

- Combined sewer overflows (CSOs): Calculated considering CSO and storm tank locations, chemical monitoring data from the Chemical Investigations Programme (CIP), rainfall intensity, and CORINE land use data.

For various points in each water body catchment, the dataset includes mean phosphate concentration (mg/L) and associated upper and lower confidence limits.

2.2 Catchment Selection

Figure 2.1 Map of benchmarking points in relation to the regions of England. The legend shows the number of measures of orthophosphate-P at each point.

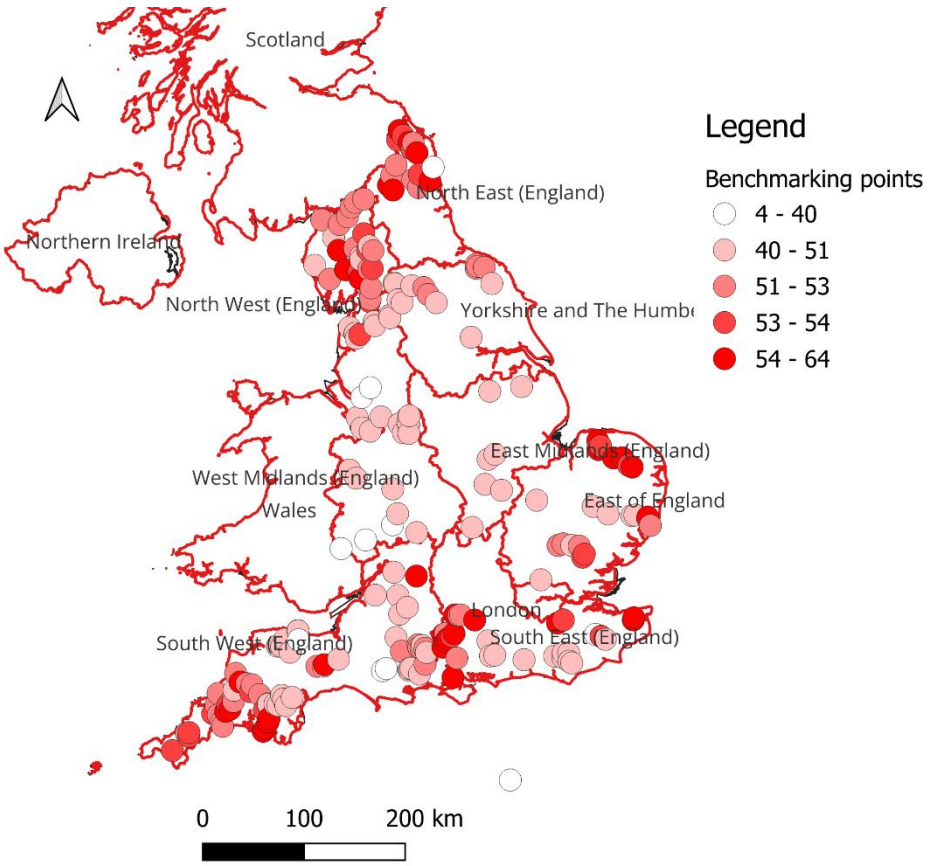


Figure 2.1 shows the positions of all points used for the RDMAI Open OrthoP model validation. These are EA monitoring points containing measures of orthophosphate-P. The validation exercise (based on EA measurements from June 2024 to July 2025 that were not included in the model training) simulated orthophosphate-P for these points and compared the simulated values to EA observations. There are limited points, and choosing specific benchmark catchments would restrict the number of points for benchmarking. In addition, there is a relative

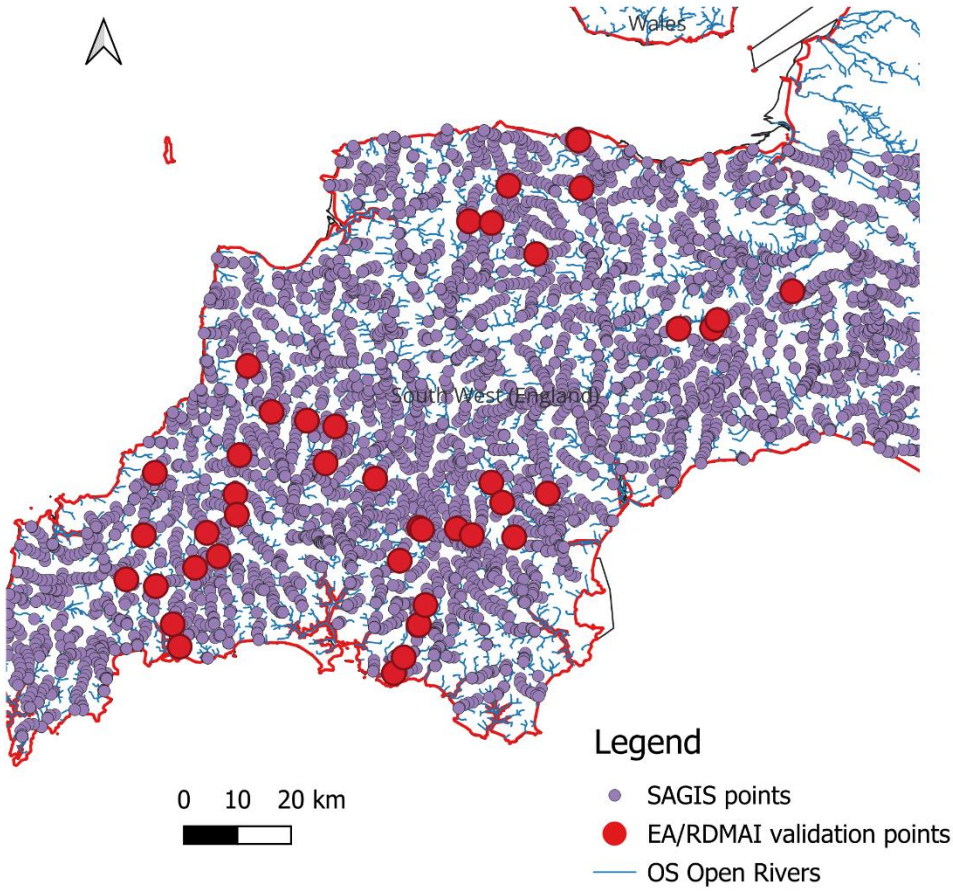
sparsity of points in the Midlands. Therefore, the decision was made to include all the points in the benchmarking and present the results at a national scale.

2.3 Benchmarking Methodology

The RDMAI Open OrthoP model was validated using data in the EA water quality archive from June 2024 to July 2025. This approach was taken because the model was trained on data in the archive prior to June 2024, and the validation dataset should be independent of the training dataset. The benchmarking of the RDMAI Open OrthoP model should similarly be on the validation dataset. The validation points are shown in Figure 2.1. During the model validation process, the RDMAI Open OrthoP model simulated orthophosphate-P measures temporally corresponding to the EA measures of orthophosphate-P for these points. The number of observations/simulations among the validation points range between 4 and 64.

The benchmarking is against an annual SAGIS orthophosphate-P measure. Therefore, the averages for the RDMAI Open OrthoP model simulations and EA observations must be calculated for each point. A threshold of 40 observations was set to get a reasonable average, i.e., points with less than 40 observations were discarded (see Figure 2.1). The UKWIR SAGIS dataset provides annual average orthophosphate-P measures for points in catchments across England. As shown in Figure 2.2, the UKWIR SAGIS dataset provides many point estimates of orthophosphate-P in catchments in England. A script was written in Python to match pairs of UKWIR SAGIS points with EA/RDMAI Open OrthoP model validation points. This matching was done purely by location – there was no common identifier between the two datasets that could be used to match pairs. A threshold distance of 50 m was chosen to match pairs of points. By connecting matched pairs with lines and plotting them in GIS, samples of matched pairs of nodes were examined to ensure that the script was working properly.

Figure 2.2 Comparison of UKWIR SAGIS and EA/RDMAI Open OrthoP model validation points for Southwest England



Within the matched pairs of points, the mean and standard deviation of orthophosphate-P for the EA/RDMAI Open OrthoP model validation points were calculated. This allowed 95% confidence limits (CLs) to be calculated as:

$$\mu \mp Z \times (\frac{\sigma}{\sqrt{n}})$$

where μ is the sample mean, σ is the sample standard deviation, n is the number of samples, and *Z* is the Z-value corresponding to a 95% confidence (1.96). The UKWIR SAGIS points also provide CLs for the mean annual estimates of orthophosphate-P concentration. The benchmarking comparison can then compare the mean and CL estimates of orthophosphate-P from the RDMAI Open OrthoP model and the SAGIS model with the mean and CLs of the EA measured orthophosphate-P.

2.4 Benchmarking Limitations

The UKWIR SAGIS model provides mean annual estimates of orthophosphate-P, whereas the RDMAI Open OrthoP model provides estimates corresponding to EA spot samples. The RDMAI

Open OrthoP model estimates then need to be averaged to compare to the UKWIR SAGIS model estimates. This approach does not account for any seasonal variation in orthophosphate-P or spikes in orthophosphate-P associated with pollution or weather events.

The RDMAI Open OrthoP model incorporates the latest available open datasets, whereas updates to the UKWIR SAGIS datasets are done periodically; therefore, the RDMAI Open OrthoP model may reflect changes to orthophosphate-P pollution sources that are not reflected in the UKWIR SAGIS dataset. In particular, the RDMAI Open OrthoP model validation set incorporates model estimates and observations for 2024/2025, whereas the UKWIR SAGIS model provides estimates based on longer-term historical data.

There are limited direct comparisons that can be made between the UKWIR SAGIS model and the RDMAI Open OrthoP model, particularly in regions with sparse validation points, such as in the Midlands (see Figure 2.1).

The matching between EA/RDMAI Open OrthoP model validation points and the UKWIR SAGIS points was done using an automated script due to the large number of points involved. Despite implementing a threshold of 50 m for matching, matched points may fall on different branches of the river waterbody and may therefore be subject to different loads of orthophosphate-P from upstream and different diluting flow.

When comparing UKWIR SAGIS and RDMAI Open OrthoP model estimates of orthophosphate-P to EA monitoring point average measures, there were limited associations that could be made in each region of England. To achieve a reasonable long-term indication of observed orthophosphate-P, only monitoring points with 40 or more measures were considered – this also allowed the calculation of reasonable confidence limits in the observed mean.

Finally, the EA measurements of orthophosphate-P, and therefore the matching RDMAI Open OrthoP model estimates, are infrequent, typically at most once or twice a month. Therefore, EA monitoring data is likely to miss spikes in orthophosphate-P associated with ephemeral pollution or weather events. Missing these spikes in orthophosphate-P in the observation record will act to push the mean downward. In contrast, the estimates of orthophosphate-P by the UKWIR SAGIS model will consider spikes due to events, and therefore, the calculated annual orthophosphate-P mean in SAGIS will be pushed up. This issue reflects a general limitation of using infrequent EA monitoring data for model training, validation and benchmarking.

2.5 Model 'Ease-of-Use'

Table 2.1 Comparison of model 'ease-of-use'

Model Outputs	SAGIS	RDMAI Model
Installation & Run Process	The SAGIS system requires expert water quality and ArcGIS users to install and run the models. The set up is not easy as it is an extensive set of files to connect to the source apportionment databases. There are associated spreadsheets to edit inputs, calibrate the model and view the model results that take time to learn to use. The user guides are comprehensive as there are a lot of detailed settings to review. Email support is available from the EA and consultants who update the SAGIS system.	The RDMAI Open OrthoP model consists of a set of Python scripts (in Jupyter notebook format), encoders for non-numerical information used in the model and the model itself. The encoders and the model are saved as pickle files, a binary type of file that can be used to save a Python object to be loaded and used at a later date. The available files, input data requirements and instructions for using the model are provided in the model's GitHub repository. While using the model is straightforward, data preparation consists of many steps and may be too involved for an inexperienced programmer. Therefore, an expert Python programmer is recommended to install the model (including required Python packages) and to adapt the scripts to prepare the available data for use. Limited instructions are available for the preparation of the input data.
Data inputs	The source apportionment data and model structures are already set up for the user. There are instructions on editing the model structure and inputs as required. This involves setting up summary statistics (e.g. mean phosphate for wastewater treatment works) or seasonal distributions (e.g. monthly diffuse phosphate loads from agriculture).	ML models require that the structure of the input data be presented exactly as it was during the model training stage. Therefore, any data will need to be pre-processed before using in the model. A list of the required determinands and their corresponding EA notation IDs is available in a README file in the model's GitHub repository. In addition to the determinands, the model uses metadata information, including sample date, sampling point name, purpose and coordinates and whether the sample was compliant. The model only recognises

Model Outputs	SAGIS	RDMAI Model
	The data updates are limited to periodic updates coordinated around the 5-year water company investment cycle and the EA requirements to use SAGIS results for the 6-year River Basin Planning and reporting cycle.	sampling points in the EA archive but can still run with data from other locations. A Python script is available in the GitHub repository that can be used to prepare data from the EA water quality archive. However, the script will need to be adapted to any other data format in order to prepare the input table.
Licensing Requirements	A license is required to use the SAGIS system. The SIMCAT software can be used with an open-source licence independently of SAGIS for individual catchments. The SAGIS report and tool are available to purchase from UKWIR for £600 (+ fees for additional data licencing). It is available through UKWIR to water companies. The Environment Agency have made some outputs available through the CaBA Data Package.	The RDMAI Open OrthoP model is open source free to use. It is built on Python using only free Python packages. Data from the EA water quality archive is generally free to use. However, costs may be incurred if using data from other sources.

2.6 Model Outputs

Table 2.2 Comparison of model outputs

Model Outputs	SAGIS	RDMAI Model
Data Format	Results from SAGIS are both in SIMCAT format (sets of text files and CSV files with georeferencing of water quality predictions and source apportionment at each model result point) and these results files are imported into the SAGIS ArcGIS Pro viewer and viewed as shapefiles. Results relevant to this comparison are long-term phosphate predictions as mean values with confidence intervals.	Running the model will result in an array containing the orthophosphate-Predictions. This can be used for producing result plots and for saving as files (CSV, Excel) as required. Additional code will need to be written to produce the desired user outputs.
Temporal Resolution	SAGIS provides a long-term annual average concentration for 38 pollutants including phosphate. Results include source apportionment, confidence limits around the mean and percentiles. Comparisons to water quality standards are also provided.	The outputs of the RDMAI Open OrthoP model will match the dates in the input data. As such the granularity of the outputs is variable.
Geographic Constraints	Model results are available for all non-tidal WFD river reaches for the UK.	The model has been trained using data from river sampling points in the EA operational catchments (England). Therefore, it is currently more suited for this data, although it has the potential for being applied elsewhere.

2.7 Time and Cost Comparison

2.7.1 SAGIS

Although the SIMCAT software is free to use, the SAGIS model version can only be used by experienced water quality modellers and requires access to ArcGIS Pro and purchase of the SAGIS tools. Costs include:

- The purchase of the latest UKWIR SAGIS report: £600 + VAT
- A license from WHS for LowFlows Enterprise for the specific geographical region of interest.
- ArcGIS Pro: £4,200/yr per named licence.

On request, the EA do provide on request truncated versions of the SIMCAT model files (.DAT) for specific river catchments that can be used with the free software. These files would include the source apportionment inputs set up by the EA to represent diffuse and point loads.

Time is needed to derive water quality baselines for the purpose of using SAGIS. A past WRc project allocated around 4 days for using SAGIS to derive water quality baselines. Using an existing SAGIS dataset, such as the UKWIR SAGIS dataset used in this benchmarking, requires a simple download. Extracting the required data from a pre-existing SAGIS dataset takes around 0.5 days.

2.7.2 RDMAI Open OrthoP Model

The RDMAI Open OrthoP Model is open source and free to use and uses open data. Data costs can be incurred if an alternative water quality dataset to the EA water quality database is used.

Processing data for input into the model requires considerable time, and an experienced Python programmer is required to understand and perform the steps for input data preparation and to run the model.

2.8 Model Performance

EA measured P, RDMAI Open OrthoP model simulated P and SAGIS simulated P were compared at a national scale for the whole of England. Comparisons were not made at the regional or catchment scale, due to the limited number of data points within many areas.

RDMAI Open OrthoP model estimates remain fairly stable at around 0.04 mg/l with increasing observed orthophosphate P concentration, until observed mean orthophosphate P concentration reaches around 0.1 mg/l, after which the RDMAI Open OrthoP model estimates start to increase. However, in absolute terms, the RDMAI Open OrthoP model estimates remain similar to the observations. In contrast, while there is a weak trend relationship between the UWKIR SAGIS model estimated orthophosphate P and mean observed orthophosphate P, the

SAGIS estimations are often far higher. Figure 2.4 shows the same data, but with a log-scale Y-axis, to show all the UKWIR SAGIS estimates.

Figure 2.3 and Figure 2.4 show that the RDMAI Open OrthoP model struggles to represent variations in orthophosphate P within the range of observed orthophosphate P concentrations. Within the range of observed orthophosphate P concentrations of 0 to 0.1 mg/l, the RDMAI Open OrthoP model estimations remain relatively static, typically ranging between 0.03 and 0.05 mg/l. The model sensitivity does seem to increase as observed orthophosphate P concentrations exceed 0.1 mg/l, although the model does seem to underpredict values consistently. The UTAG standards for orthophosphate, depending on elevation and alkalinity, range between 0.04 and 0.12 mg/l, 0.028 and 0.07 mg/l, 0.087 and 0.173 mg/l, and 0.75 and 1.00 mg/l for High, Good, Moderate and Poor status sites, respectively. Therefore, it is reasonable to assume that the RDMAI Open OrthoP model is not sufficiently sensitive to inform orthophosphate P status for High and Good sites and could perhaps have more applicability at Moderate and Poor sites (i.e. where orthophosphate P concentrations are typically greater than 0.1 mg/l). However, only 15 of 114 compared sites had an observed annual mean concentration great than 0.1 mg/l, so further data would be needed to understand whether the RDMAI Open OrthoP model could perform accurately at these locations.

Figure 2.5 shows X-Y scatter plots of UKWIR SAGIS and RDMAI Open OrthoP model estimations verses mean EA observed orthophosphate P for the paired points. The imposed linear regression lines show that overall, the UKWIR SAGIS model overestimates EA measured orthophosphate P concentration, while the RDMAI Open OrthoP model overall underestimates. The marginally greater R^2 of the UKWIR SAGIS model is due to a more consistent trend relationship with measured EA orthophosphate P concentration, whereas the estimates of the RDMAI Open OrthoP model remain static with increasing measured orthophosphate P concentration, and only start increasing beyond 0.1 mg/l.

While the orthophosphate P estimates of the UKWIR SAGIS model are often orders of magnitude higher than observations, it should be considered that the infrequency of EA measures of water quality (typically around 1 per month at a site) leads to spikes in orthophosphate P being missed. Therefore, these high UKWIR SAGIS estimates are not necessarily incorrect.

Figure 2.3 shows a comparison between mean EA measured orthophosphate P concentration and that simulated by the RDMAI Open OrthoP model and the UKWIR SAGIS models. The Y-axis is cut off at 0.4 mg/l since the extreme high UKWIR SAGIS estimates for some points distort the graph scale. The results show that the RDMAI Open OrthoP model estimates remain fairly stable at around 0.04 mg/l with increasing observed orthophosphate P concentration, until observed mean orthophosphate P concentration reaches around 0.1 mg/l, after which the RDMAI Open OrthoP model estimates start to increase. However, in absolute terms, the RDMAI Open OrthoP model estimates remain similar to the observations. In contrast, while there is a weak trend relationship between the UWKIR SAGIS model estimated orthophosphate P and mean observed orthophosphate P, the SAGIS estimations are often far higher. Figure 2.4 shows the same data, but with a log-scale Y-axis, to show all the UKWIR SAGIS estimates.

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While the orthophosphate P estimates of the UKWIR SAGIS model are often orders of magnitude higher than observations, it should be considered that the infrequency of EA measures of water quality (typically around 1 per month at a site) leads to spikes in orthophosphate P being missed. Therefore, these high UKWIR SAGIS estimates are not necessarily incorrect.

Figure 2.3 Comparison of orthophosphate-P concentrations simulated by the RDMAI Open OrthoP model and the UKWIR SAGIS model with EA water quality measures for England for 2024/2025. Site IDs shown on the X-axis are ordered according to ascending mean EA observed orthophosphate P concentration.

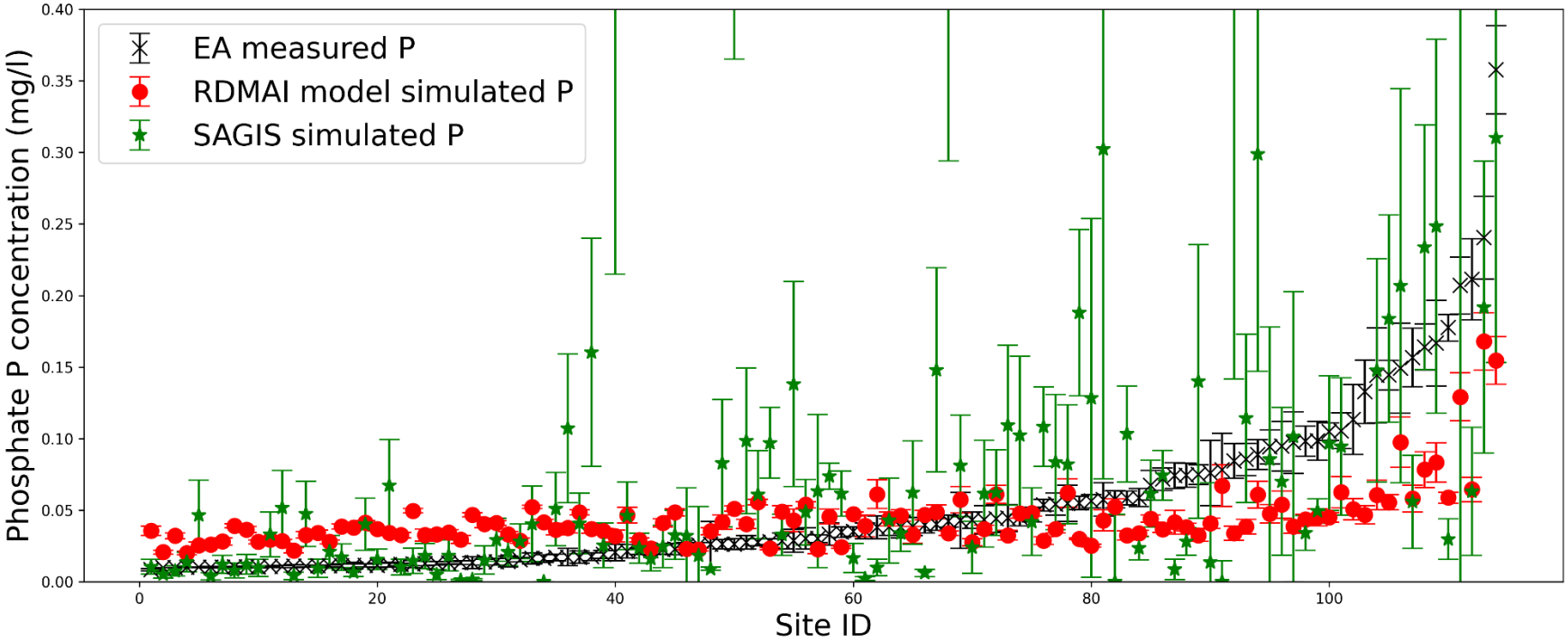


Figure 2.4 Comparison of orthophosphate-P concentrations simulated by the RDMAI Open OrthoP model and the UKWIR SAGIS model with EA water quality measures for England for 2024/2025. Site IDs shown on the X-axis are ordered according to ascending mean EA observed orthophosphate P concentration. The Y-axis is shown on a log scale.

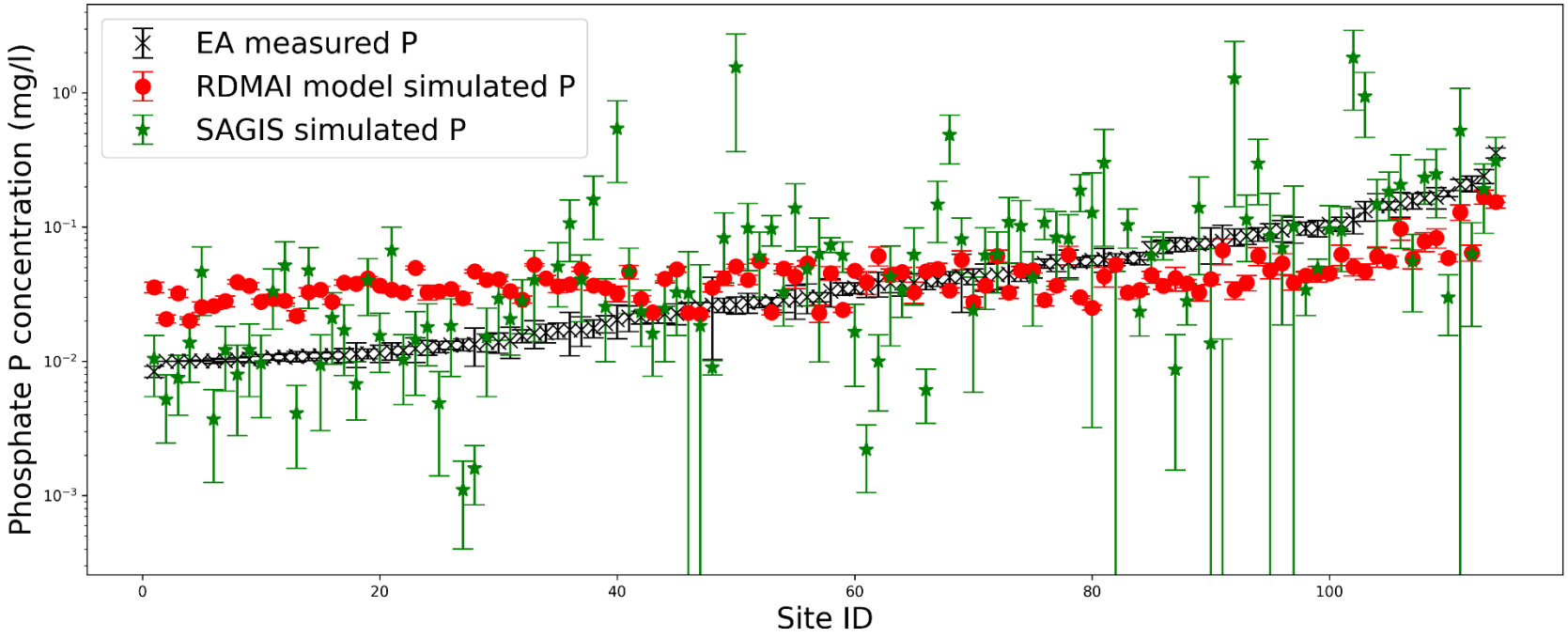
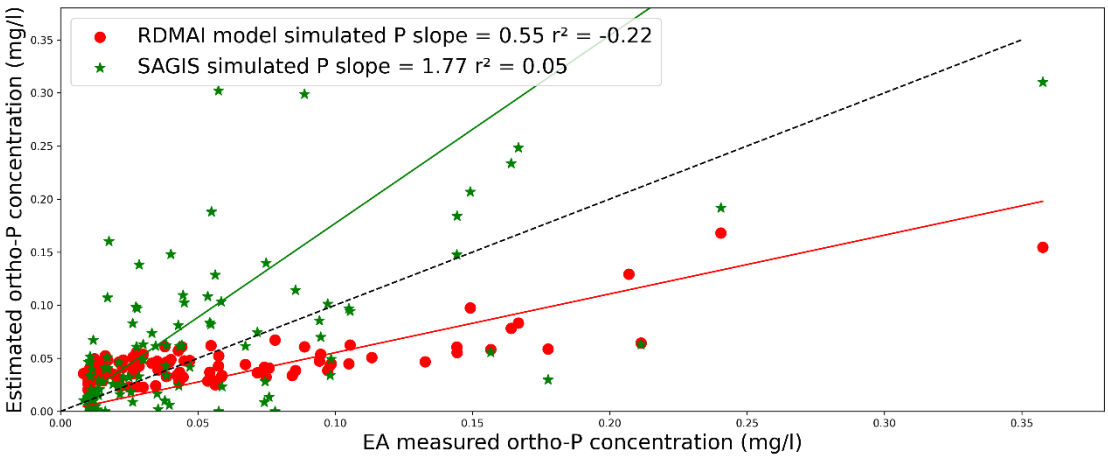


Figure 2.5 X-Y scatter plots of model estimated versus mean EA measured orthophosphate P concentration for paired points, with fitted regression lines. A 1:1 line is shown for reference (black dotted line). As both axes are kept to the same scale for ease of graph interpretation, some higher UWKIR SAGIS estimates fall outside of the plot limits



2.9 Model Performance Summary Statistics

It should be stressed that the RDMAI Open OrthoP model incorporates the latest available open datasets, and the benchmarking is done against EA measured orthophosphate P for 2024/2025. In contrast, since updates to the UKWIR SAGIS datasets are done periodically, the UKWIR SAGIS dataset may not reflect recent improvements to orthophosphate P, such as due to improved sewage treatment or reduced CSO spills. This limitation should be kept in mind when considering the benchmarking results.

To compare the relative performances of the two models (UKWIR SAGIS and RDMAI Open OrthoP model), the Relative Root Mean Square Error was calculated as a comparison between the mean model estimates and the mean EA measures, across the paired points (see Table 2.3). The RDMAI Open OrthoP model has a much lower RRMSE compared to the UKWIR SAGIS model, indicating that the RDMAI Open OrthoP generally performed better than the UKWIR SAGIS model at representing an annual mean measured orthophosphate-P. Looking at the plots in Figure 2.3 and Figure 2.4, while there is a broad trend relationship between the UKWIR SAGIS estimations and observed orthophosphate P concentrations, the SAGIS estimations are often orders of magnitude higher. This is highlighted by the high RRMSE of the UKWIR SAGIS model shown in Table 2.3. However, given the low frequency of EA measures of orthophosphate P, it is likely that spikes in concentration due to ephemeral pollution events are often missed. Therefore, the UKWIR SAGIS estimates are not necessarily incorrect.

As shown in Figure 2.3 and Figure 2.4 and further illustrated in Table 2.3, the RDMAI Open OrthoP model estimates were much closer to EA observed concentrations of orthophosphate P. However, the RDMAI Open OrthoP model estimates remained stable and did not react to

increasing observed EA measurements of orthophosphate P until a concentration of around 0.1 mg/l was exceeded. Given orthophosphate P concentrations below this threshold affect the eutrophic status of a river reach, these results imply that the RDMAI Open OrthoP model could perhaps have more applicability to moderately or highly eutrophic sites, if further model development and testing was undertaken. The results of the benchmarking reflect the conclusions of the model validation. The validation concluded that model was more suitable at predicting orthophosphate concentrations under 'normal' conditions but struggled at sites with extreme high or low orthophosphate readings. These model characteristics are reflected in Figure 2.3 of the benchmarking, where the model over-estimates extreme low observed orthophosphate concentrations as predictions remain stable at around 0.04 mg/l. While the model predictions do increase at sites with higher observed orthophosphate concentrations, they underpredict overall.

Table 2.3 A summary of the relevant performances of the UKWIR SAGIS model and RDMAI Open OrthoP model when benchmarked against EA measurements of orthophosphate-P

Number of sites compared	RRMSE		R^2		
	RDMAI Open OrthoP model	UKWIR SAGIS	RDMAI Open OrthoP model	UKWIR SAGIS	
114	80%	512%	−0.22	0.05	

3. Benchmarking against INCA-P

An additional benchmarking exercise was conducted using data and simulation results obtained from INCA-P[3] (INtegrated CAtchment model of phosphorus). It should be noted that the INCA-P model was not provided, only the simulation results. INCA-P provides outputs in timeseries format, similar to the RDMAI Open OrthoP and therefore, offers a different comparison approach to SAGIS.

INCA-P is part of the INCA suite, a set of water quality models that dynamically simulate land and instream processes for different water quality determinands, used extensively in studies around the world. INCA-P models simulate the behaviour of dissolved and particulate phosphorus, as well as stores and fluxes of water and sediment[3].

The INCA-P model is generally considered easy to use, although it requires intermediate to advanced modelling skills and GIS knowledge to set it and to interpret the simulation results.

Input requirements include:

- Daily time series of hydrologically effective rainfall (HER), soil moisture deficit (SMD), precipitation and mean air temperature. The first two can be obtained from the PERSiST hydrology model, which is designed to work with models in the INCA suite.
- Land use and soil types to define land classes.
- Inputs of fertiliser, manure and sewage.
- Observations for model calibration and validation, including:
    - Instream discharge and water chemistry data, including different phosphorus forms as required.
    - Knowledge of nutrient uptake in harvest, for validation of soil processes.

The INCA-P models can provide the following outputs:

- Daily time series of flows, water quality concentrations at sub-catchment outlets along the river.

[3] L.A. Jackson-Blake et al. The INtegrated CAtchment model of phosphorus dynamics (INCA-P): Description and demonstration of new model structure and equations. Environmental Modelling & Software, v. 83, pp. 356-386, 2016.

- Profiles of flow or water quality along the river for any day.
- Cumulative frequency distributions of flow and water quality at selected sites.
- Daily and annual pollution loads for all land uses and all processes.

INCA-P is a semi-distributed model, meaning it simplifies the physical catchment into land use classes within each sub-catchment and simulates the processes within each class separately, instead of accounting for the full spatial distribution of the land features. This reduces the data and computational requirements. As long as the catchment can be simplified as required, INCA-P can be used independently of the geographical location.

A licence for INCA-P may be made available free of charge for academic use, but the documentation requests that Water Resources Associates be contacted before the model is used for commercial use. Pricing is not publicly available but can be obtained through enquiry from Water Resources Associates. The models are open source, and the code can be accessed on their GitHub repository. It is recognised that licencing a product that is open source seems counterintuitive, but nevertheless, that is what the model documentation states.

3.1 Catchment selection

Data for the benchmarking was available for an unpublished case study in the Tarland Burn[4], a sub-catchment of the River Dee in northeast Scotland. The study was carried out by the James Hutton Institute and the University of Reading and aims to explore the effectiveness of measures to reduce nutrient concentrations in streams, under both current and future climate and land use, by simulating different pollutants including phosphorus.

For this benchmarking exercise, monitoring data was available for the monitoring point at Coull and included mean daily flow and concentrations of suspended sediment (SS), total dissolved phosphorus (TDP), total phosphorus (TP), particulate phosphorus (PP), soluble reactive phosphorus (SRP), nitrate ($NO3^-$-N) and ammonium ($NH4^+$-N), collected between 2000 and 2010. Sampling for water chemistry took place at a weekly frequency between 2000 and the end of 2003, with some daily sampling during rainfall events. Daily samples were then collected between February 2004 and June 2005. After June 2005, infrequent irregular sampling has taken place, with fewer determinands.

INCA-P and INCA-N (nitrogen) models were used to simulate flows and concentrations of the parameters listed above. The results of the simulations were also made available for the benchmarking.

[4] Leah Jackson-Blake and Andrew Wade. Exploring future water quality at catchment scale – case study in the Tarland Burn, northeast Scotland. DEFRA integrated modelling project, Case Study 4.

3.2 Benchmarking Methodology

The RDMAI Open OrthoP is built on a set of water quality determinands using EA's specific notation for each determinand. The Tarland Burn data that matched with determinands used in the RDMAI Open OrthoP are listed in Table 3.1. With the exception of SRP (orthophosphate-P) all determinands in the table were used as model inputs for benchmarking.

Table 3.1 Determinand notation matches

Tarland Burn determinand	EA determinand notation	EA determinand name	EA determinand description
suspended sediment (SS)	0135	Sld Sus@105C	Solids, Suspended at 105 C
total phosphorus (TP)	0348	Phosphorus-P	Phosphorus, Total as P
soluble reactive phosphorus (SRP)	0180	OrthoP	Orthophosphate, reactive as P
nitrate ($NO3^-$-N)	0117	Nitrate-N	Nitrate as N
ammonium ($NH4^+$-N)	0111	Ammonia(N)	Ammoniacal Nitrogen as N

The Tarland Burn data was processed so that it was in the required format for use as inputs for the RDMAI Open OrthoP. The data processing included:

- Adapting existing Python scripts (in Jupyter notebook format) from the model GitHub repository.
- Renaming the determinands to match the model input table.
- Using saved encoders in the repository to convert categorical/non-numeric data into labels suitable for modelling. This included metadata such as name of the sampling point and whether each sample was compliant.
- Generating additional variables created for the model to find patterns.
- Add season information based on dates.

Once the input data table was prepared, it was fed into the model to generate the predictions of orthophosphate-P.

The Tarland Burn report provided with the data did not include model performance metrics for the SRP simulation results. In order to compare models, some of the metrics considered in the RDMAI Open OrthoP performance assessment were applied to both models as follows:

- RMSE (Root Mean Square Error): quantifies the average magnitude of the prediction error.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:
y_i = observed values
$\hat{y}_i$ = predicted values
n = number of observations

- NRMSE (Normalised Root Mean Squared Error): expresses RMSE relative to the observed data, making the error dimensionless and allowing the comparison between datasets with different scales. In this case, the normalisation was done in relation to the range of the observed data. This constrains the values between 0 and 1 for interpretability.

$$NRMSE = \frac{RMSE}{y_{max} - y_{min}}$$

Where:
y_{max} = maximum observed value
y_{min} = minimum observed value

- R^2 (Coefficient of Determination): measures the proportion of the variance in the data that is explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Where:
$\bar{y}$ = mean of the observed data
Other terms as defined in the equations above.

3.2.1 Benchmarking Limitations

The RDMAI Open OrthoP model was developed using 45 water quality determinands. In contrast, only four matching determinands were available for the Tarland Burn data, meaning that all other determinands were entered as zeros in the model input table. This restricts the predictive capacity of the model, as it expects certain relationships between the determinands to be taken into account. Zero values are not realistic for some environmental data, resulting in unpredictable model behaviour.

The RDMAI Open OrthoP was trained using data measured in sampling points within the EA operational catchments, which are restricted to England. Information about the location of these

sampling points is encoded in the input data used for training the model. The Tarland Burn is a catchment in Scotland with location unknown by the model and with characteristics that may be outside of the data used to train the model.

The Tarland Burn catchment is a mixture of intensive agriculture, rough grazing, forestry and moorland[4]. Daily phosphate concentrations are generally difficult to simulate in agricultural catchments and require lag features, (e.g. to take into account the time it takes to reduce the available soil phosphorus store after reductions in fertilizer/manure inputs) (L. Jackson-Blake, personal communication, 15 October 2025). However, the RDMAI Open OrthoP does not include lag features, which may compromise its performance on this data set.

Benchmarking against the INCA-P model should recognise that the INCA-P model incorporates information about weather, soil moisture deficit and flow[4], which are known to influence the orthophosphate-P concentrations. These variables were not used in the RDMAI Open OrthoP model. As a result, differences in the predictive performance may reflect not only the modelling approach or skill, but also the effect of these variables on the orthophosphate-P concentrations.

3.3 Benchmarking Results

Both INCA-P and the RDMAI Open OrthoP provide outputs as time series. The INCA-P model provides daily concentrations of each determinand simulated, while the RDMAI Open OrthoP provides estimates of orthophosphate-P at the same frequency as the input data.

Figure 3.1 shows the time series of the observed orthophosphate-P measurements compared to the INCA-P simulations and the RDMAI Open OrthoP predictions. The INCA-P simulation results provide continuous, daily values, while the RDMAI Open OrthoP provides a prediction for each date where the input data is available. The figure shows that both models have difficulties predicting the orthophosphate-P at some points, which could be expected given the known difficulties to predict orthophosphate-P in this type of catchment. The RDMAI Open OrthoP, however, has a general tendency to overestimate the orthophosphate-P values, with some underestimated values below zero, which is an impossibility. Negative predictions are possible because the LightGBM algorithm is unconstrained and free to extrapolate outside the physically valid range. In this particular case, this is likely to be caused by the inherent limitations of the model (see Section 3.2.1), which can result in the model extrapolating results due to limited input information.

Figure 3.1 Comparison of the RDMAI Open OrthoP predictions with the observed values and the INCA-P simulations

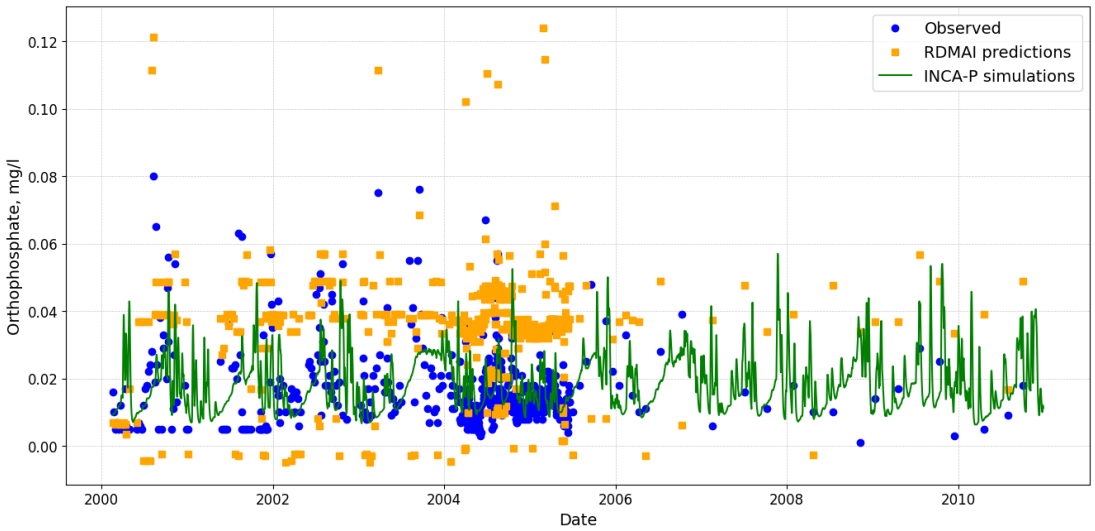


Figure 3.2 shows a comparison of the distributions of the orthophosphate-P on all three datasets. The data in this figure is transformed into a log-scale because the observed orthophosphate-P data is skewed (i.e. includes many values near the limit of detection) and also include spikes. Transforming the data in this case, makes the structure of the distributions clearer and easier to compare. Figure 3.2 also shows the tendency of the RDMAI Open OrthoP to overestimate the orthophosphate-P concentrations. It should be noted that the log transformation excludes the negative predictions in the RDMAI Open OrthoP model, as negative values cannot be log-transformed. However, the negative predictions account for only 4.9% of all predictions in this dataset.

Figure 3.2 Comparison of the orthophosphate-P distributions (on log-scale)

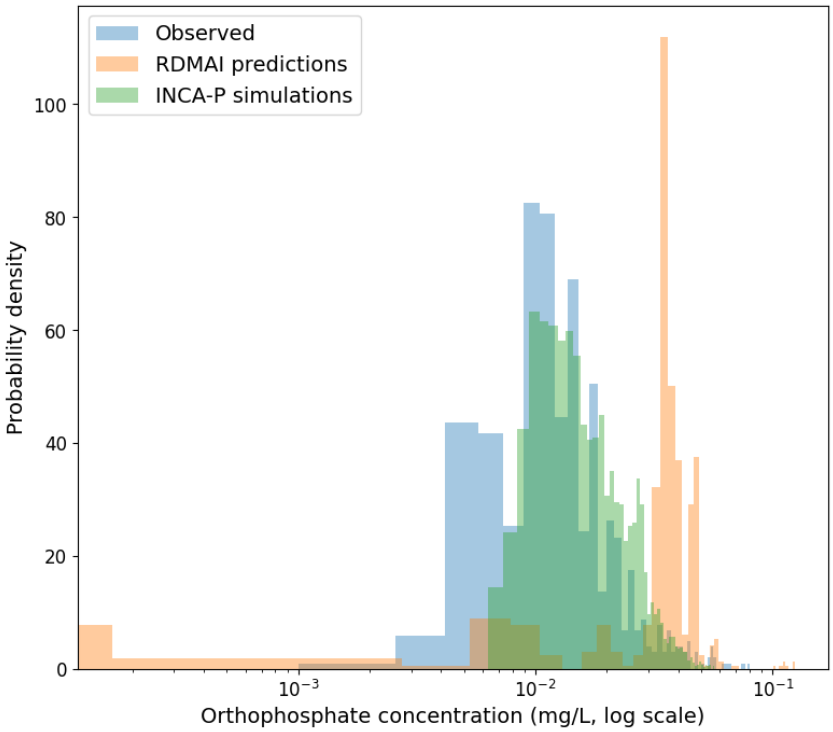
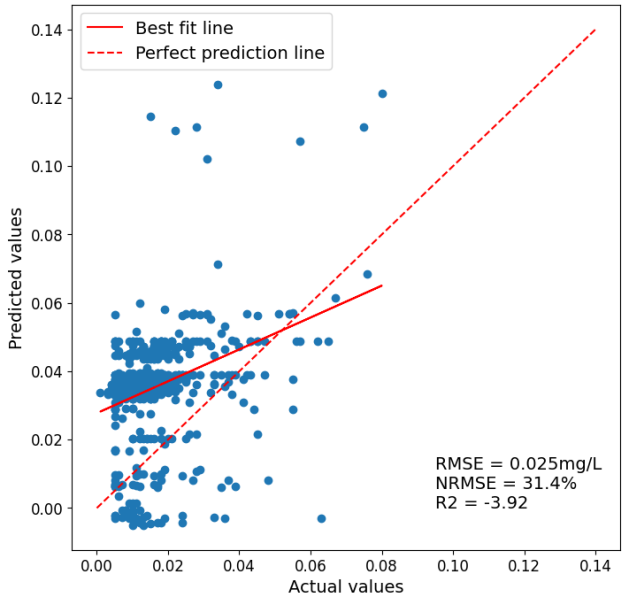


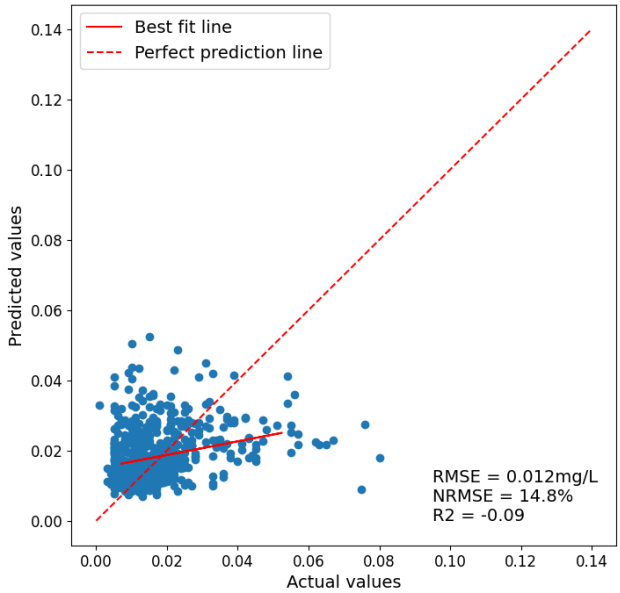
Figure 3.3 shows the actual vs predicted plots resulting from both models and includes the values of the metrics calculated. Each plot shows the best-fit line based on the model results (solid red) and a perfect prediction line (dashed red). The same scale was used in both plots for comparison. The RDMAI Open OrthoP plot shows considerable deviation from the perfect prediction line, mainly overpredicting (as shown in Figure 3.1). The relatively high NRMSE (31.4%) confirms that the prediction errors are large relative to the observed values. The INCA-P simulation results show lower RMSE and NRMSE (0.012mg/L and 14.8%, respectively), indicating that the predictions are closer to the observed values in absolute terms than the RDMAI Open OrthoP (RMSE = 0.025mg/L). Both models resulted in negative R^2 (-3.92 for the RDMAI Open OrthoP and -0.09 for the INCA-P model), indicating that the models explain substantially less variance than the mean of the observed data, with the INCA-P model still showing better performance. It should be noted, however, that the INCA-P model, being process-based, is generally not calibrated just to maximise fit (as is the case for ML models), but is also built to reflect the physical and biogeochemical processes. In this case, a low or negative R^2 does not necessarily indicate poor model performance.

Figure 3.3 Actual vs predicted plots. A) RDMAI Open OrthoP predictions; B) INCA-P simulations.

A) RDMAI Open OrthoP predictions



B) INCA-P simulations



4. Benchmarking Conclusion

4.1 Summary

A summary table of key model characteristics can be found in the Summary, at the start of the report.

4.2 Conclusion

The RDMAI Open OrthoP model estimates of orthophosphate-P were in general more representative of EA observations at a mean annual level than the estimates by the UKWIR SAGIS model. However, the benchmarking indicates that the RDMAI Open OrthoP model may not be sufficiently sensitive to represent orthophosphate P concentrations in sites with a High or Good status, according to the UTAG orthophosphate P standards, and can be applied with more confidence to sites of a Moderate or Poor status. The shortcomings of such a temporally lumped benchmarking must be recognised, as it does not account for seasonal or short-term variations in orthophosphate-P due to pollution or rainfall events. In addition, the benchmarking was done against EA measures of orthophosphate P for 2024/2025, and while the RDMAI Open OrthoP model incorporates the latest data, recent improvements to orthophosphate P may not be reflected in the UKWIR SAGIS dataset. The general limitations of benchmarking against infrequent EA measures must also be recognised, since these measures often miss spikes in pollution. The benchmarking result against the UKWIR SAGIS model should be viewed in the context of these limitations. Nevertheless, this benchmarking indicates that the RDMAI Open OrthoP model may be more suitable for estimating orthophosphate-P concentrations in sites experiencing higher orthophosphate-P concentrations (above 0.1 mg/l), when comparisons to observations are made on an annual mean basis. However, overall, the RDMAI Open OrthoP model tends to underestimate measured orthophosphate-P for concentrations exceeding 0.1 mg/l and produces relatively static estimations for observed orthophosphate-P concentrations less than 0.1 mg/l.

The benchmarking against the INCA-P model highlights that the RDMAI Open OrthoP model has relatively limited accuracy when the availability of determinands is very sparse compared to what the model was trained on and performs best with as much information as possible. In addition, running the model using data from Scotland when it was trained using data solely from England was also expected to impact its performance. Therefore, there is an opportunity to expand the model beyond the EA's operational catchments so that it recognises characteristics in other geographical areas. Additional model improvements could be achieved, for example, by re-training the model with a log-transformed response to reduce negative predictions.

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